Exercise 2 SQL Fundamentals (Aggregate Functions & Groping)

1) Count() the number of employees

```
SELECT COUNT(id) AS number_employees
FROM 'workspace'.'default'.'employees';
```

number_employees
10

2) Total salary paid to employees in the IT department

```
SELECT SUM(Salary) AS total_salary
FROM 'workspace'.'default'.'employees'
WHERE department = 'IT';
```

total_salary
220 000

3) Calculate the AVG salary in the HR department

```
SELECT AVG(Salary) AS average_salary
FROM 'workspace'.'default'.'employees'
WHERE department = 'HR';
```

average_salary
49500

4) Find the highest and the lowest salary in the company

```
SELECT MAX(Salary) AS highest_salary
       MIN(Salary) AS lowest_salary
FROM 'workspace'.'default'.'employees';
```

highest_salary	lowest_salary
62000	48000

5) Group employees by department and total salary in each department

```
SELECT department,
        SUM(salary) AS total-salary
FROM 'workspace'.'default'.'employees'
GROUP BY department;
```

department	total-salary
IT	220000
HR	99000
Finance	119000
Marketing	105000

6) Count how many employees in each city

```
SELECT city,
        COUNT(id) AS number_employees
FROM 'workspace'.'default'.'employees'
GROUP BY city;
```

city	number_employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7) Group employees by department and calculate avg salary in each department, order the result in decending order.

```
SELECT department,
        AVG(salary) AS average_salary
FROM 'workspace'.'default'.'employees'
GROUP BY department
ORDER BY average_salary DESC;
```

department	average_salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

8) Find the department where the total salary paid exceeds 100 00 (HAVING CLAUSE)

```
SELECT department
        SUM(salary) AS total_salary
FROM 'workspace'.'default'.'employees'
GROUP BY department
HAVING total_salary > 100000;
```

department	total_salary
IT	220000
Finance	119000
Marketing	105000

9) List cities where more than one employee works, ordered by the number of employees in decending order

```
SELECT city,
        COUNT(id) AS number_of_employees
FROM 'workspace'.'default'.'employees'
GROUP BY city
HAVING number_of_employees >1
ORDER BY number_of_employees DESC;
```

city	number_of_employees
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10) Find the department with the highest average salary

```
SELECT department,
        AVG(salary) AS average_salary
FROM 'workspace'.'default'.'employees'
GROUP BY department
ORDER BY average_salary DESC
LIMIT 1;
```

department	average_salary
Finance	59500